

AI / ML Learning Roadmap

Sequential order • suggested time • main source per topic

Computer Vision

#	Topic	Time	Main Source
1	ResNet	2 days	"Deep Residual Learning for Image Recognition" (He et al., 2015)
2	DenseNet	1.5 days	"Densely Connected Convolutional Networks" (Huang et al., 2017)
3	MobileNet	1.5 days	"MobileNets: Efficient CNNs for Mobile Vision" (Howard et al., 2017) + MobileNetV2 paper
4	EfficientNet	2 days	"EfficientNet: Rethinking Model Scaling" (Tan & Le, 2019)
5	ConvNeXt	2–3 days	"A ConvNet for the 2020s" (Liu et al., 2022)
6	Vision Transformer (ViT)	3 days	"An Image is Worth 16x16 Words" (Dosovitskiy et al., 2020)
7	Transfer Learning	1–2 days	CS231n notes (Stanford) — "Transfer Learning" section
8	Object Detection (fundamentals)	4 days	"R-CNN" → "Faster R-CNN" papers, or d2l.ai "Object Detection" chapter
9	YOLO	3–4 days	YOLOv1 paper (Redmon et al., 2016) + Ultralytics YOLO docs for modern versions
10	Segmentation + U-Net	3–4 days	"U-Net: Convolutional Networks for Biomedical Image Segmentation" (Ronneberger et al., 2015)

NLP → LLM

#	Topic	Time	Main Source
11	Tokenization + Embeddings	2 days	"word2vec" paper (Mikolov et al.) + Hugging Face "Tokenizers" docs
12	BERT + GPT	3–4 days	BERT paper (Devlin et al., 2018) + GPT-1/2 papers (Radford et al.)
13	LLM Architecture	3–4 days	"Attention Is All You Need" (Vaswani et al., 2017) + Jay Alammar's "Illustrated Transformer"
14	Fine-tuning + LoRA	3–4 days	"LoRA: Low-Rank Adaptation" (Hu et al., 2021) + Hugging Face PEFT docs
15	RAG + Vector DB	4–5 days	"Retrieval-Augmented Generation" (Lewis et al., 2020) + LangChain / LlamaIndex docs
16	Agents + Tool Calling	3–4 days	"ReAct: Synergizing Reasoning and Acting" (Yao et al., 2022) + Anthropic/OpenAI tool-use docs

Generative AI

#	Topic	Time	Main Source
17	VAE	shared 3–4 days w/ GAN	"Auto-Encoding Variational Bayes" (Kingma & Welling, 2013)
18	GAN	shared 3–4 days w/ VAE	"Generative Adversarial Networks" (Goodfellow et al., 2014)
19	Diffusion Models	4–5 days	"Denoising Diffusion Probabilistic Models" (Ho et al., 2020) + Lilian Weng's diffusion blog post

Multimodal AI

#	Topic	Time	Main Source
20	CLIP + VLM	3–4 days	"Learning Transferable Visual Models From Natural Language Supervision" (Radford et al., 2021)
21	Multimodal AI (general / Multimodal LLM)	3–5 days	Flamingo paper (Alayrac et al., 2022) or LLaVA paper — good overview of modern multimodal LLM design

Rough total: ~62–74 days if done sequentially full-time, though many topics (e.g., the ResNet family, or VAE/GAN/Diffusion) can be studied in parallel since they don't have strict dependencies within their own group — only the CV → NLP → GenAI → Multimodal ordering across groups really matters, since Multimodal AI leans on both vision and language foundations.